



Principles of Crown Preparation

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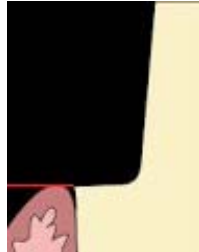
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42

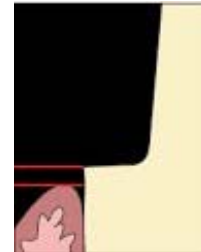
Margin/Finish Line Placement



Sub-gingival



Crestal



Supra-gingival



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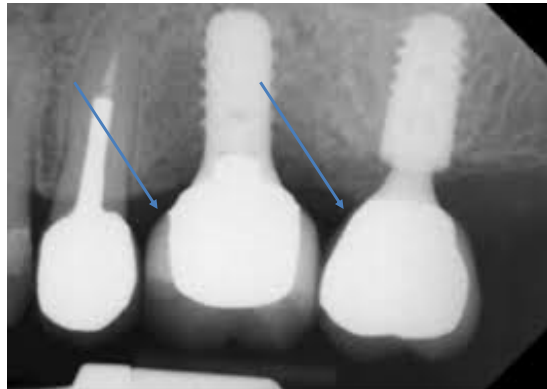
43

In the previous lecture we discussed the 3 positions that a restorative dentist can place their margin:

- 1) Supra-gingivally
- 2) At the gingival Crest
- 3) Sub-gingivally

We will now go into more depth on this topic and the factors that will determine where you place your margin.

Emergence Profile



Emergence Profile is the contour of the tooth or restoration, such as a crown on a natural tooth or implant abutment as it relates to the adjacent tissues. To break this down, it is the shape of the tooth as it emerges from the gum tissues. Depending on the location of your crown you will need to change the location of your margin so that you can recreate the proper emergence profile.

Supra or Crestal Margin Placement



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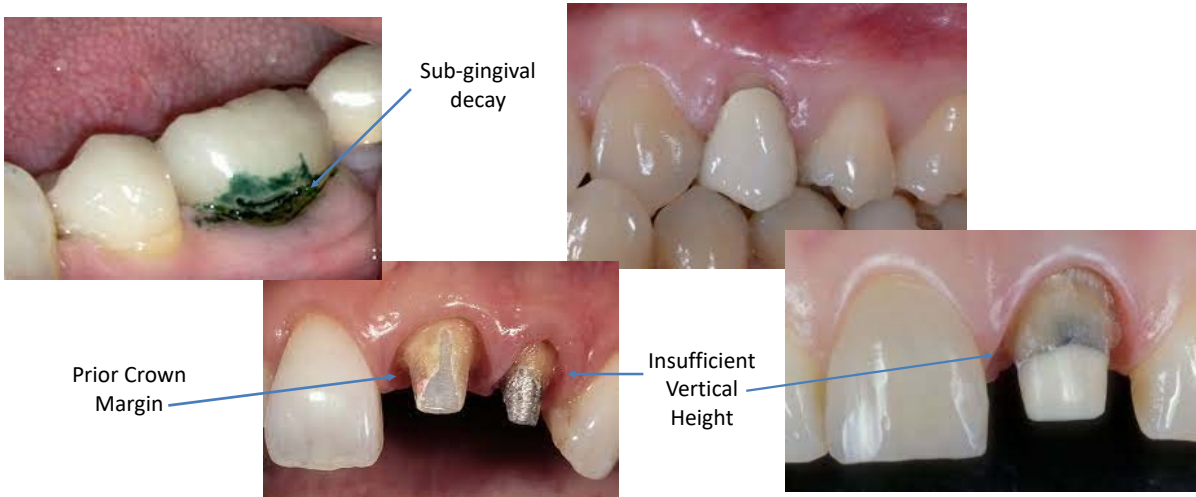
45

With the advent of strong porcelain materials like lithium disilicate or zirconia movement of the margin above the gums has become more common place. These materials are more esthetic and subsequently we do not have to hide the transition of crown to tooth.

There are a few other advantages to a supra-gingival margin.

- 1) They are easy to finish without trauma to the tissues
- 2) Easy for the patient to keep it plaque free
- 3) Easier to impress with less potential for soft tissue damage
- 4) The restoration can easily be evaluated at the time of placement and at recall appointments

Sub-gingival Margin



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46

There are times when you have to move your margin sub-gingivally. One instance is when a prior crown already has a margin there. You will be forced to place your margin below the gum line. At times existing decay will extend sub-gingivally, this will require a margin below the gums to establish the margin on sound tooth structure. Certain crown materials like Porcelain Fused to Metal crowns, PFMs, can be very unaesthetic at the margin due to the gray metal. In the aesthetic zone, we have to hide this and many times the only way to do that is extending our margin below the gum line. Another indication would be that the proximal contact for the adjacent tooth extends apically to the level of the gingival tissues. At times, the root surface will have sensitivity that you can not control by other means so you will need to extend you margin apically to these areas to help. Sometimes you will need to change the axial contour of a tooth. These are usually for things like a partial denture. Finally, the clinical crown exposed may be too short to achieve proper retention form and resistance form, which will be discussed later in this lecture. In these situations, dropping or extending your margin sub-gingivally can give you more tooth structure to work with.

There are some challenges to a sub-gingival margin. These include ensuring that the finish line is smooth, challenges in making an impression and hard to check the final marginal fit.

Hygiene



The margin areas of a crown are inherently rougher than virgin tooth structure. This will harbor more bacterial causing inflammation. Margins at the crest of the tissue or above will be easier for the patient to clean. When a margin is placed below the crest of the gingival it can be a source of inflammation for the tissues. When trying to clean below the gum line, a toothbrush bristle will only go about 0.5 mm subgingival. Floss can go about 2.5mm and a water flosser can clean down to 4mm.

Gingival Inflammation



Sub-gingival margins can and will cause localized inflammation of the tissues around the crown. The inflammation can range from mild/subclinical to severe. The severe inflammation can present as swelling of the tissues, redness, tenderness to touch, and bleeding which can lead to bone loss. The patient's inflammatory response will be dictated by the patient's overall gingival health and their biotype. The gingival inflammation can also result in gingival recession. Generally, a shallower healthy sulcus is less likely to have the gingiva recede while a deeper sulcus has a greater chance of recession.



Controlling Inflammation

- Marginal Depth
- Quality of the marginal fit
- Smoothness of the restoration material

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49

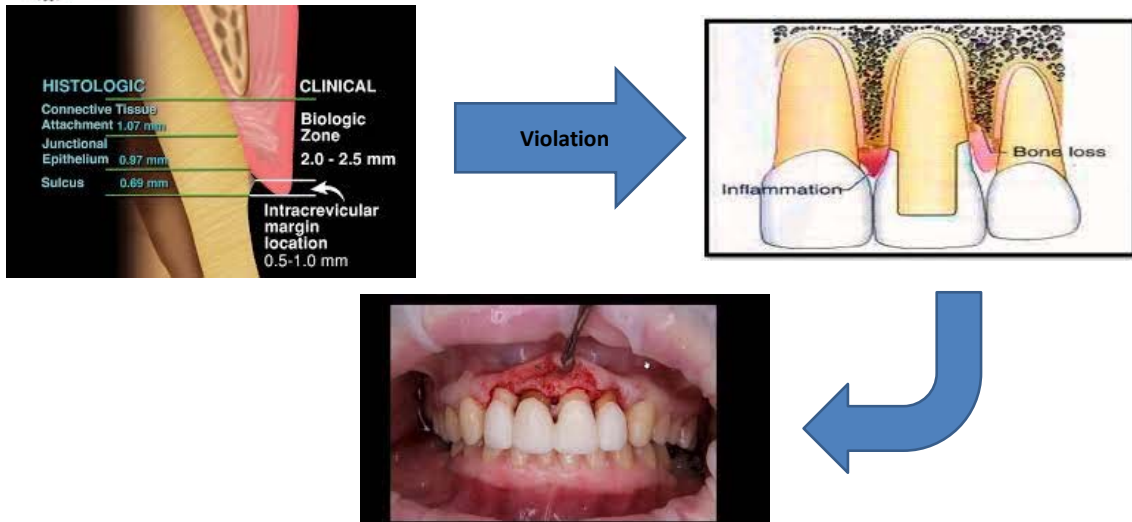
Dentists can help control the inflammation in three ways.

The first is the marginal depth. The deeper the margin goes the more likely the patient will not be able to clean it and the higher likelihood of a poor fit. The general rule for a sub-gingival crown margin is no more the $\frac{1}{2}$ the sulcus depth. In addition, it is recommended that you have at least 3mm of attached gingiva.

The second way to control the inflammation is to have a high quality of a marginal fit. The tighter the fit the less likely to harbor bacteria. If the margin is open more than 200 microns the more likely that the margin will harbor bacteria thus causing inflammation.

The third way is the smoothness of the restoration. The rougher the restoration will again harbor bacteria causing inflammation. We want the restoration to be as smooth as possible. This might mean polishing areas of adjustments or have a high-quality glaze on the crown.

Biological Width



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50

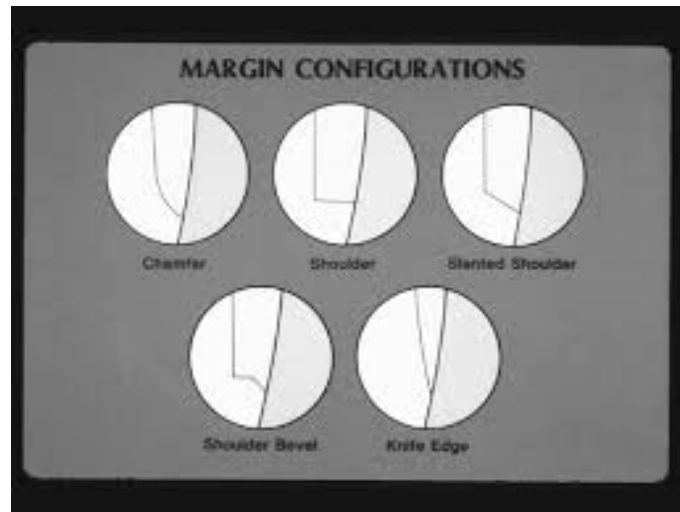
When a sub-gingival margin is placed it is possible to violate Biologic Width. This has been or will be discussed in more detail in the Perio course. When biological width is violated there will be an inflammatory response that will eventually lead to bone loss. This is the body's way of re-establishing biological width, but it will not be in a controlled manner.

If there is a violation in biological width with your crown margin, there are two methods of correction: surgical or orthodontic extrusion. The surgical correction is a form of periodontal surgery that will move the bone and gingival tissues apically in a controlled manner. If periodontal surgery is performed you will need to wait 3 months for posterior teeth and 6 months for anterior teeth before you set your new marginal placement. With orthodontics, the tooth is extruded to expose more tooth structure and moves the margin away from the bone.

So, if you are going to have a violation how do you prevent this. Just as with a violation that exists you can perform periodontal surgery in the form of crown lengthening or perform orthodontic extrusion. As will be discussed in the perio course, you need to be conscious of how the tissues will heal on the neighboring teeth in the Anterior. This can cause asymmetric tissue position and teeth appearing longer which can be unaesthetic to the patient. Additionally, many periodontists will want you to place your margin subgingivally first, so they know where to set the bone and tissues. Good communication

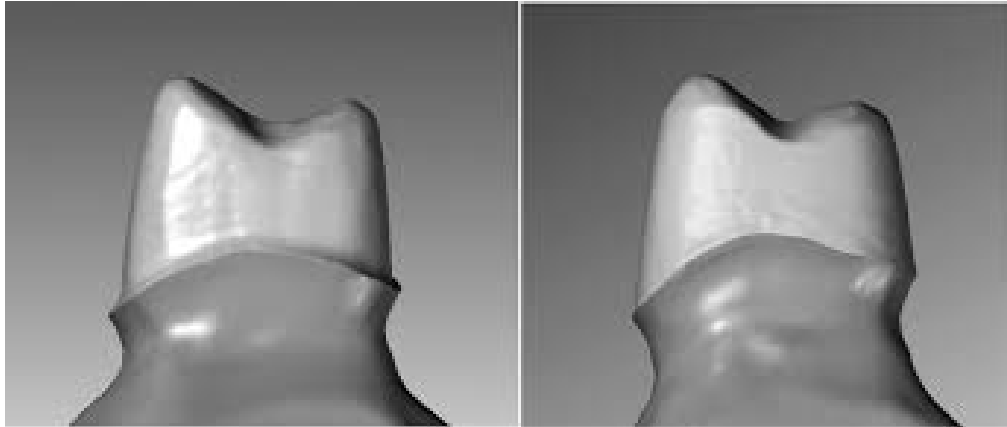
with the periodontist is a must.

Marginal Geometry



Once you have determined the location of your margin and finish line, you now have to determine the shape of your margin which is called the marginal geometry.

Margin Geometry



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52

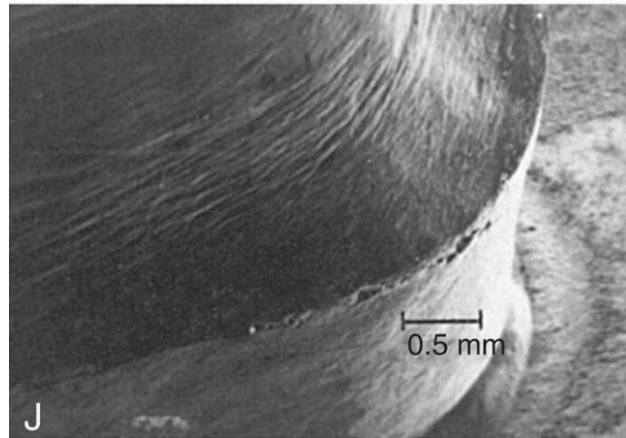
The shape the margin takes is dictated by the crown material that you are using.

There are general guidelines for the design or geometry of the margin.

- 1) The design should not overextension or unsupported enamel at the cavosurface angle
- 2) Should facilitate easy identification on the impression or scan
- 3) Should be a distinct boundary for waxing and finishing
- 4) Should have sufficient bulk for the restorative material used
- 5) Should conserve tooth structure if items 1-4 have been met

Typical geometries used for crowns are 1) Chamfer 2) Shoulder 3) Beveled Shoulder 4) Knife edge.

Chamfer



Pictures from Contemporary Fixed Prosthodontics 6th Edition

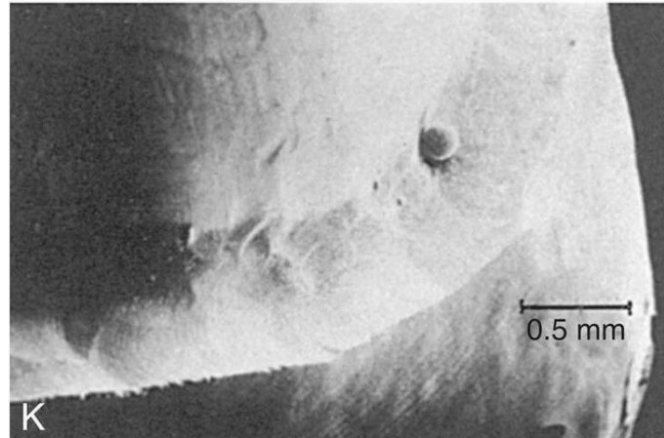
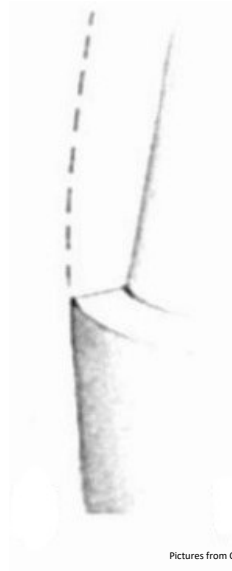
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53

A chamfer is created when a rounded end diamond sunk into the tooth structure up to halfway of the diameter of the tip of the bur. This will create a rounded transition from the margin to the axial wall. If the bur is taken more than half of the diameter it has a tendency to create unsupported enamel. In the cross section this will look like a ski jump or skateboard ramp. These areas of unsupported enamel need to be removed. This can be done with your chamfer diamond or an end cutting diamond bur. This marginal design can be used for most crown materials. We will be using a modification of this marginal styles.

Shoulder



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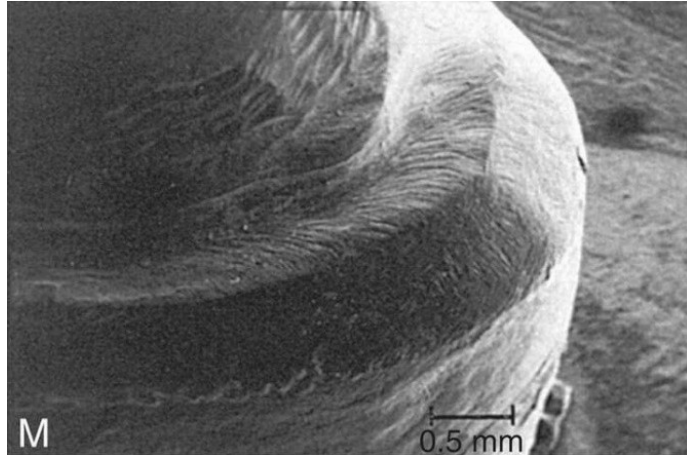
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54

A shoulder margin has more of a sharp angle at the transition to the axial wall. The diamond will not be rounded at the tip and will leave a blunt edge at your finish line. These are good for PFM crowns where porcelain will be at the margin and metal will be at the transition to the axial wall.

Beveled Shoulder



Pictures from Contemporary Fixed Prosthodontics 6th Edition

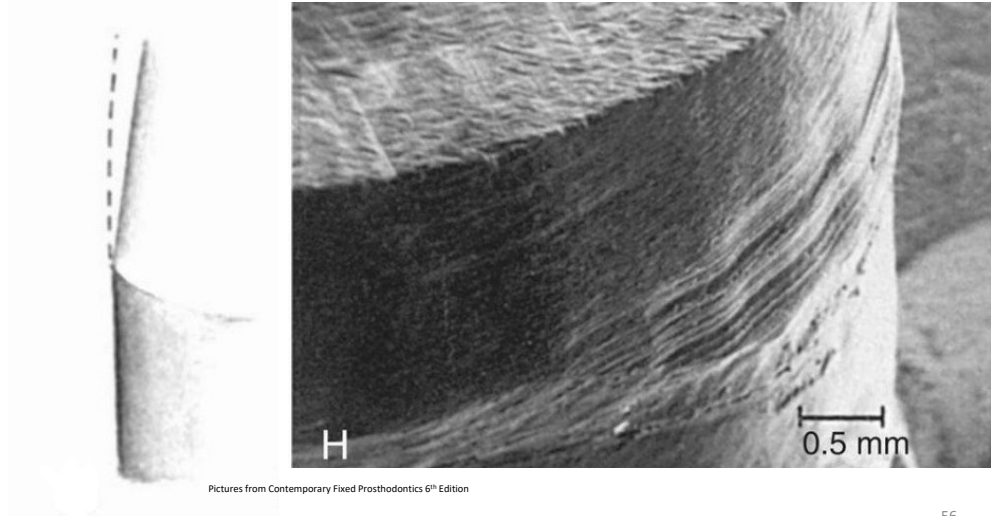
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55

A beveled shoulder is the same as a shoulder with one difference. In this marginal design the finish line is beveled. You can use this marginal design when doing a gold crown or having a metal margin for a PFM crown.

Knife Edge



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56

A knife edge margin is as the name describes very thin with minimal reduction at the finish line. This type of margin was typical for a gold crown. Over time as new materials were introduced the marginal design was slow to change with it on the operator side. Doctors would use this marginal design for PFM or porcelain crowns. This causes overcontoured restorations. The overcontour will create a food and bacterial trap inducing gingival inflammation. It is very important to reduce the axial wall enough so that there is proper space for the crown material.

Margin Design	Advantages	Disadvantages	Indications
Feather edge	Conservative of tooth structure	Does not provide sufficient bulk	Bonded ceramic restorations
Chisel edge	Conservative of tooth structure	Location of margin difficult to control	Occasionally on tilted teeth
Beveled	Removes unsupported enamel, allows finishing of metal	Extends preparation into sulcus if used on apical margin	Facial margin of maxillary partial-coverage restorations and inlay/onlay margins
Chamfer	Distinct margin, adequate bulk, easier to control	Care needed to avoid unsupported lip of enamel	Cast metal restorations, lingual margin of metal-ceramic crowns, porcelain laminate veneers, ceramic restorations
Shoulder	Bulk of restorative material	Less conservative of tooth structure	Facial margin of metal-ceramic crowns, ceramic restorations
Sloped shoulder	Bulk of material, advantages of bevel	Less conservative of tooth structure	Facial margins of metal-ceramic crowns
Beveled shoulder	Bulk of material, advantages of bevel	Less conservative, extends preparation apically	Facial margin of posterior metal-ceramic crowns with supragingival margins

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This is a table from the textbook: Contemporary Fixed Prosthodontics 6th Edition . It reviews the positives and negatives for each style of marginal design.

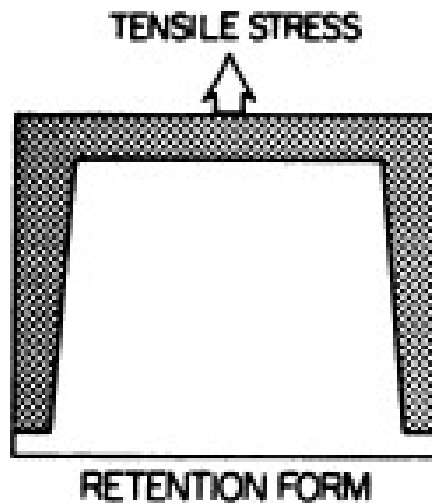


Mechanical Considerations

- Retention Form
- Resistance Form
- Deformation Prevention

Now that you have prepared a few typodont teeth for crowns, there are mechanical considerations that need to be discussed because they will affect the success of your restoration . These are Retention Form, Resistance Form and Prevention of Restoration Deformation. We will discuss each of these individually.

Retention Form



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59

Retention Form is defined as the design or quality of a preparation that prevents the restoration from becoming dislodged by forces parallel to the path of placement or insertion. Only caries or fracture of the crown cause more failures to poor retention form.

There are 5 factors that need to be considered to produce good Retention Form:

- 1) Magnitude of the dislodging force
- 2) Geometry of the preparation
- 3) Roughness of the fitting surface of the restoration
- 4) Material Being cemented
- 5) Film thickness and properties of the luting agent

#5 will be discussed more in another lecture.

Magnitude of the Dislodging Force



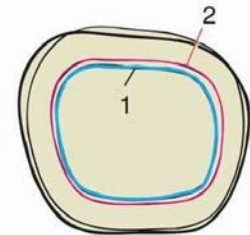
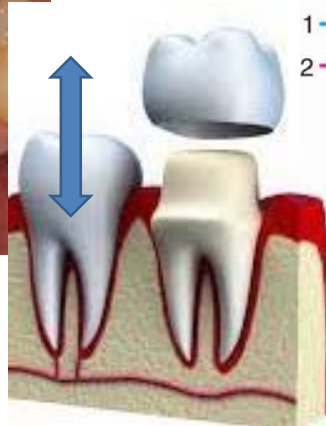
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60

The magnitude of the dislodging forces when it comes to retention form are generally less than the forces of seating or tilting. The greatest forces generally come from sticky foods like candy. The overall magnitude of the force will be determined by the elevator muscles, the stickiness of the food, the surface area and the texture of the surface area. With stickier foods like candies such as “Sugar Daddies” or “Dots” the elevator muscles will need to generate more opening force which could dislodge a crown. Also, the larger the occlusal surface the more surface area the sticky foods have to adhere to causing the muscles to work harder. Additionally, if the surface of the crown is rough the food will adhere better and thus require more force when opening.

Prep Geometry



Pictures from Contemporary Fixed Prosthodontics 6th Edition

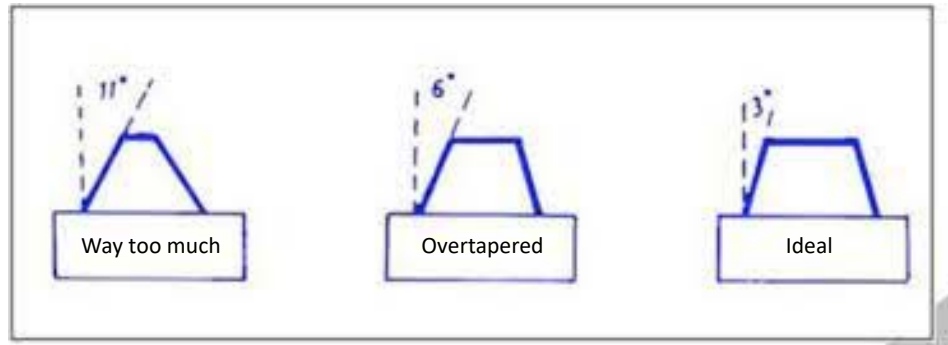
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61

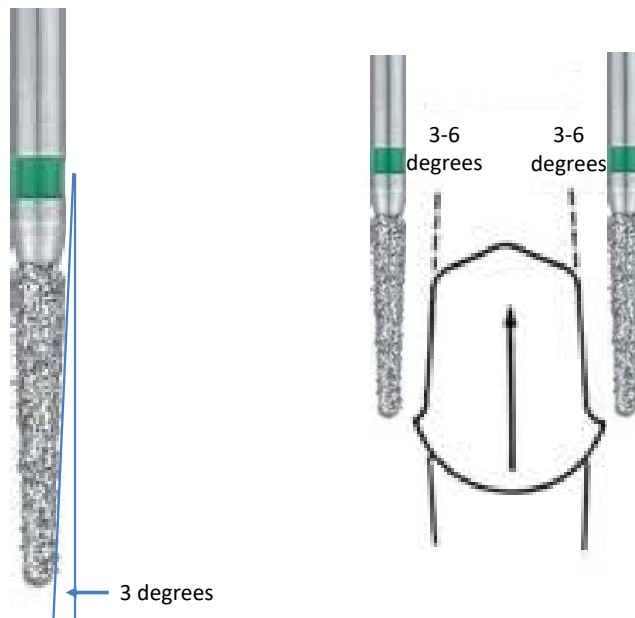
The geometry of the preparation is of a cylindrical shape with the walls at a constant angle. The ideal geometry would be a cylinder with no convergence of the walls. This would be very hard to seat, and most likely undercuts would prevent the seating of the crown. In the past, the geometry was much more important to retention than the luting agent when it came to preventing dislodgement. The better the geometry the greater the friction and thus better retention. When the geometry dictates a single path of insertion there will be better retention. The path of insertion is the pathway that a crown will go to fully seat. When you look down at a crown prep as in the lefthand picture you will see a distinct pathway that the crown will need to take in order to seat. The tighter this pathway is the greater the retention. We need to be careful that the pathway is not too narrow because the crown will not seat at all. The righthand picture shows with a consistent preparation you will have a good pathway of insertion.

Taper



To review: Taper is defined as the convergence of two opposite facing external walls of a crown preparation as viewed in a given plane. Taper can have a significant effect on the retention form of a crown. As the taper grows there will be more freedom of movement of the crown which will reduce the retention. This is demonstrated in the images on the slide.

Tapered Diamond



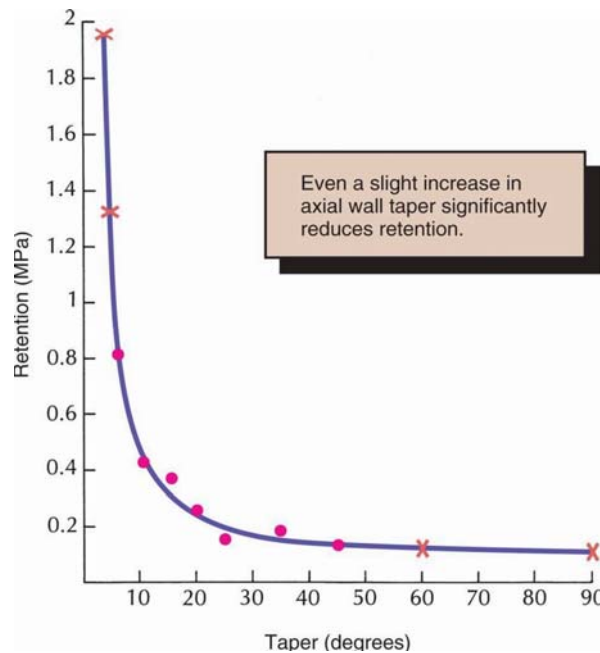
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63

A tapered diamond will have 3 degrees of taper. If this bur is held at a constant angle for the two opposing walls the ideal 6 degrees of taper will be achieved.

Taper vs Retention



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64

Taper has a hyperbolic relationship with retention. As the taper increases the retention rapidly decreases initially then after a certain point the more you increase the taper the less effect that it will have on the retention.

Good vs Poor Taper



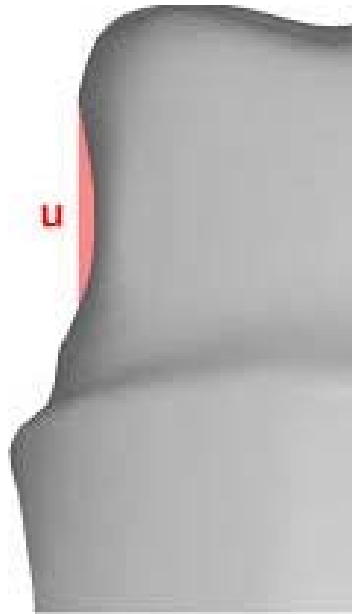
Good Taper



Poor Taper

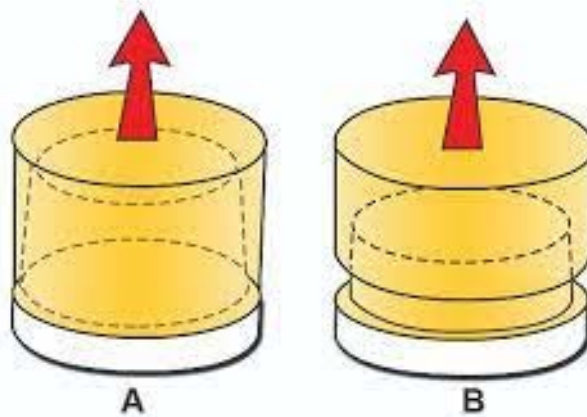
These pictures are examples of good taper and poor taper. The left side shows prepared teeth that have a more ideal 6 degree taper. The right side shows excessive taper from the buccal to the lingual. This will make it easier to seat the crown but your retention will come mainly from your cement and not the preparation.

Undercuts



When discussing retention and taper it is important to discuss undercuts. An undercut is any irregularity in a wall of a prepared tooth that prevents the withdrawal or seating of a crown. This is demonstrated in the picture on the left. The red area depicts an undercut. If a crown has this area incorporated into the internal walls, it would prevent the crown from seating or from being removed. This is something that a dentist has to be conscious of when you are evaluating your preparation. To evaluate a preparation for an undercut, you will want to view the prep from the occlusal or incisal and you should be able to see all of the external walls. If you can not see a portion of the wall, then an undercut exists and it will need to be corrected. To correct an undercut, use your tapered diamond on the wall to smooth the shape until the area of undercut is gone. Generally, undercuts will be closer to the marginal area and are caused by improper angulation of the bur when preparing the axial wall.

Surface Area



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67

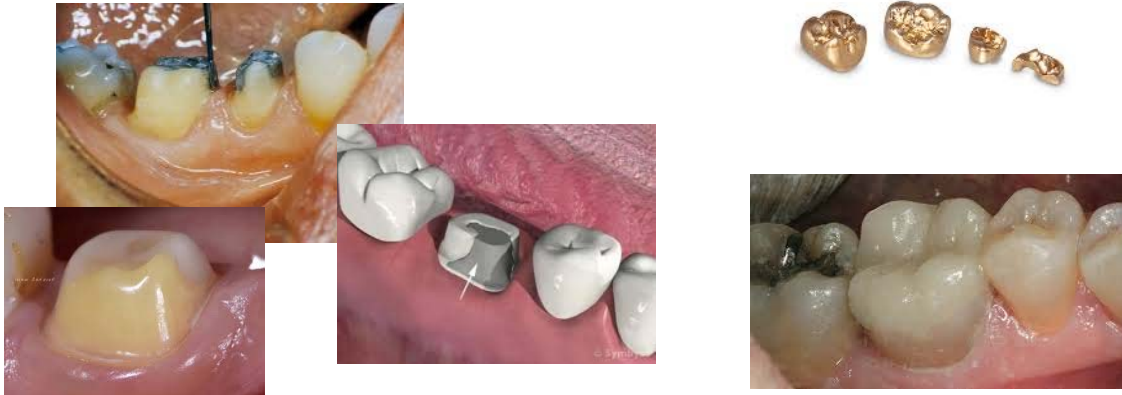
Another factor in the geometry that affects retention is surface area. The greater area of tooth that can interact with a crown the better the retention you will achieve. With this being true a longer axial wall will have better retention than a shorter one. Additionally, a molar crown will have better retention than a premolar. This is due to greater surface area that the crown must slide along. This is demonstrated in the picture. Figure A has greater surface area and thus will have greater retention than figure B.

Roughness



The next factor that affects retention is the surface texture of the internal portion of the crown. When the surface is too smooth you will see a failure of adhesion at the junction between the luting agent and the crown. It is important that the internal surface is roughened to generate greater retention of that crown. Each material you will need to roughen differently. Metal and zirconia crowns are roughened with air abrasion while a porcelain crown is roughened with an acid etch. Typically, air abrasion is done with aluminum oxide particles while hydrofluoric acid is used to etch the porcelain.

Materials Being Used



The materials that you use for a crown and in a build-up can also affect the retention. Certain materials will adhere better to composite or amalgam. It is important to pay attention to what type of crown you are using and what materials exist on the tooth.

Luting Agents

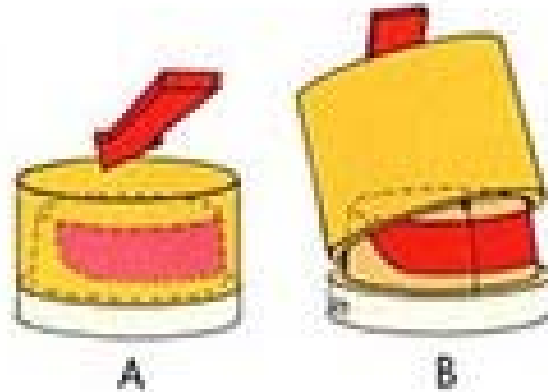


There are all types of luting agents available on the market, from traditional cements to resin bonded cements. Generally, the resin bonded cements will have the greatest retention, but the material interface will have a say in this.

Factor	Greater Retention	Lesser Retention
Taper	Parallel	Excessive
Surface area	Large	Small
Type of preparation	Molar complete crown	Intracoronary restoration
Surface texture	Rough	Smooth
Film thickness	Effect uncertain	
Luting agent	Adhesive resin	Zinc phosphate

This is table 7-4 from Contemporary Fixed Prosthodontics by Rosenstiel. It shows how different factors effect your retention.

Resistance Form



Pictures from Contemporary Fixed Prosthodontics 6th Edition

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72

The next mechanical consideration is Resistance Form. Resistance form is defined as a feature of a tooth preparation that enhances the stability of a restoration and resists dislodgement along any axis other than the path of insertion. This consideration can be more important than retention form. The geometry of the preparation will play a key role in this. We need to shape our preparation to prevent rotation of our crown because occlusal forces are not just down the long axis of the tooth. Many times, there are forces applied to a crown that are not in the long axis of the tooth and can cause a crown to rotate which can generate dislodging forces and eventually the crown will be dislodged.



Resistance Form

- Wall Taper
- Preparation Diameter
- Preparation Height

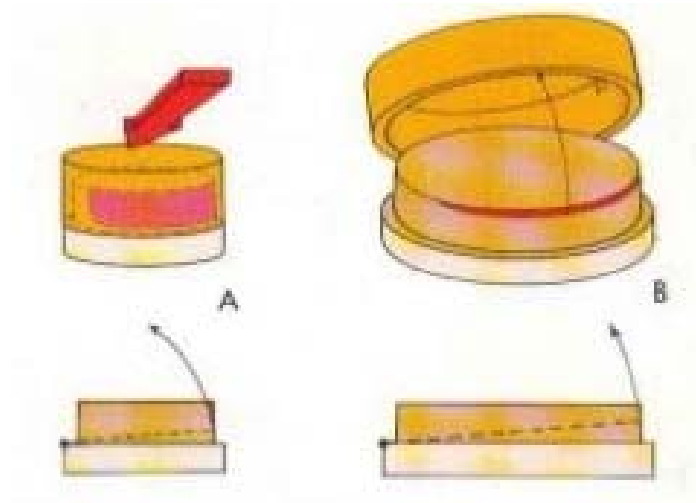
The resistance form that you create in your preparation is a function of 3 things. They are Wall Taper, Preparation Diameter and Preparation Height.

Wall Taper



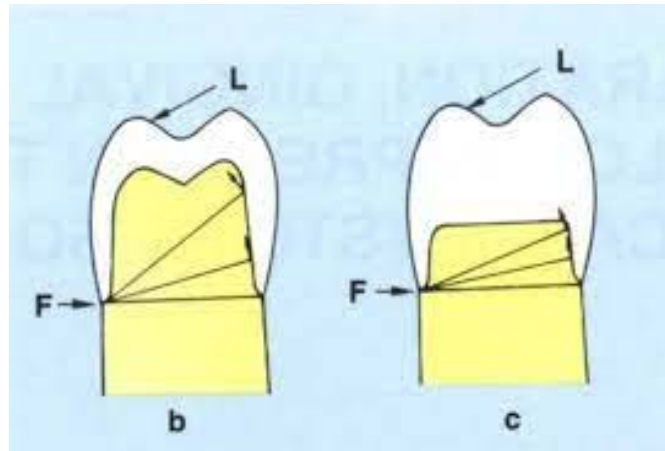
As with retention form the taper of the preparation wall has large effect. As the taper increases the ability of the tooth to resist dislodgement greatly decreases.

Preparation Diameter



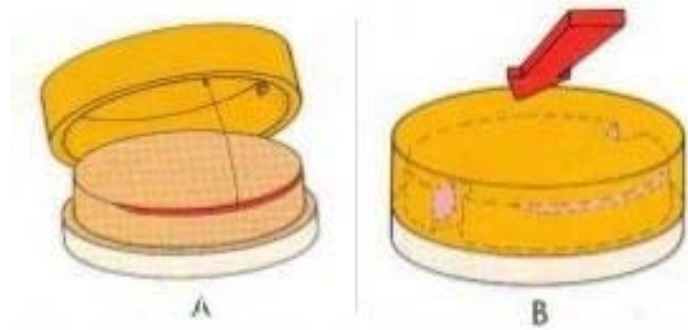
The preparation diameter is just like the diameter of a circle. As your diameter increases your resistance to tipping will decrease. This would be like comparing the diameter of a premolar to that of a molar. The retention of the molar crown would be reduced in comparison to the premolar.

Preparation Height



Preparation Height is the length of your axial wall. As the height of that wall decreases so will the resistance to dislodgement. An example of this would be a molar with a short clinical crown vs. a molar with a longer clinical crown. Ideally, for a molar you want an axial height greater than 3.5mm.

Boxes and Grooves



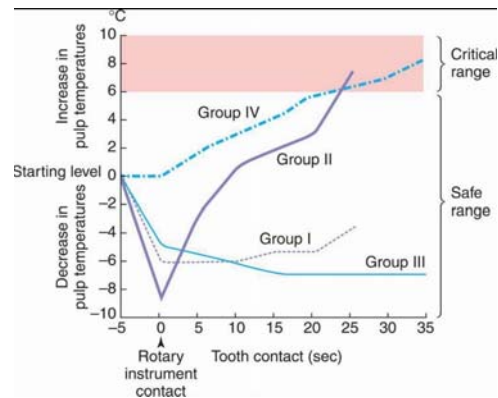
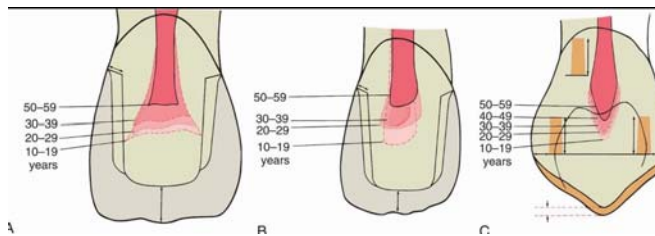
There are preparation design elements that can be added to your preparation that will enhance your resistance form. These are boxes and grooves. These types of elements were typically used for gold inlay/onlays/crowns or with PFMs that have a metal coping. They are not as commonly used with porcelain restorations. This is because the boxes and grooves needed sharp angles to help provide greater resistance, but we do not want that with porcelain. Rounded boxes are typically used with partial coverage restorations and can be placed in a full porcelain crown preparation. It is vital that the box or groove is placed in sound tooth structure. If it is placed on a build up or weakened tooth structure, this will be a site of stress and ultimately failure.

Preventing Deformation



The last mechanical consideration is preventing deformation. Now this is deformation in the restorative material. Whether it be gold, metal or porcelain or a combination of these, they will deform under occlusal forces. We need to design our preparation so that we can resist this deformation. The first step is to choose the proper material for the task at hand. If you are using gold it is important to use the proper type of gold because regular gold is too malleable and soft. Next, we want to make sure our reduction for the occlusal and axial walls are appropriate. If we do not reduce enough the induced deformation will cause a failure in the restoration. The occlusal reduction for a metal occlusal is 1.5mm and all ceramic is 2.0mm. The axial reduction for gold is 0.5mm, PFM 1-1.2mm and ceramic 1-1.5mm.

Damage to the Pulp



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79

Damage to the pulpal tissues can come from many different elements. This first is direct exposure of the pulpal tissues. As you can see in the figure on the Left side as we age the size of the pulp chamber reduces in size. We need to account for the size and shape of the pulp chamber before we begin our preparation to ensure that our preparation design will not expose the pulpal tissues. As with all preparation of tooth structure, heat generation can be a problem. In a study by Zach et al, they found that using water cooling decreased the temperature of the pulp and kept it in the safe range. Additionally, a sharp diamond bur will cut more efficiently than a dull diamond creating less heat. Lastly, since we have now removed most if not all of the enamel the pulpal tissues can be more susceptible to chemical or bacterial irritation.

All of these different types of irritants can cause pulpal death. It is important that you discuss this with a patient during your consent.

Conservation of Tooth Structure



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80

As you can see when you prepare a full coverage crown you will remove a lot of tooth structure. This can be upwards of 60-75% of the tooth structure. Since this the case, we want to be conscious of this and try and conserve as much tooth structure as possible. Our preparation must conform with mechanical principles and esthetic demands. When we are conservative, we can reduce the harmful effects to the pulpal tissues because the remaining dental thickness is inversely proportional to the pulpal response. This means with less dentin there is a greater pulpal response.

How can we conserve tooth structure? The following are a few ways to conserve tooth structure.

- 1) If possible, do a partial coverage restoration
- 2) Prep the tooth with a minimum practical taper
- 3) Prep the occlusal portion of the tooth so that it follows the anatomical shape of the tooth
- 4) Prep the axial surface so that a maximum residual tooth structure is retained
- 5) Select your marginal geometry that is conservative and conforms to the principles of preparation for your selected materials
- 6) Avoid unnecessary apical extension of the preparation.